

Wavelet Analysis and Applications

1. GENERAL

1.1 TITLE:Wavelet Analysis and Applications

1.2 COURSE NUMBER (if known)::DE.MA513.16

1.3 CREDITS:: 3-0-2(11)

1.4 SEMESTER-OFFERED::IX

1.5 PRE-REQUISITES::

1.6 Course Committee Member:

2. OBJECTIVE::

The course aims to provide concepts of wavelets and their applications to various disciplines.

3. COURSE TOPICS::

UNIT I: Fourier Transform :

Properties of Fourier Transform, inversion of Fourier Transform, relation between Fourier Transform and DFT (Discrete Fourier Transform), Convolution-an introduction, Fourier Transform in $L^2(\mathbb{R})$, Parseval formula, Plancherel relation, Poisson summation formula, summation kernels arising from Poisson summation. (10 lectures)

UNIT II: Wavelet Transform

The continuous wavelet transform, inversion formula for wavelet transform, inversion formula, properties of Wavelet Transform, discrete Wavelet Transform, frames, frame operator and its properties, orthonormal Wavelets, definition of multiresolution analysis and examples, properties of scaling functions and orthonormal Wavelet bases, construction of Wavelets, (15 lectures)

UNIT III: Applications:

Basic aspects of Fourier series, definition of Fourier series, examples of Fourier series, Fourier series of real functions, pointwise convergence of Fourier series, further aspects of convergence of Fourier series, Fourier sine series, Fourier cosine series, convergence of Fourier sine & cosine series.

Some application of Fourier series, Heat equation, the wave equation, Schrodinger's equation of free particles, Filters used in signal processing, designing filter, convolution and point spread function.

(14 lectures)

4. READINGS

4.1 TEXTBOOK::

4.2 *REFERENCE BOOKS::

1. A First Course in Wavelets with Fourier Analysis, Albert Boggers, Francis J. Norcowich, Prentice Hall.

2. Wavelet Transform and Their Applications, LokenathDebnath, Birkhauser.

3. Ten Lectures on Wavelets, Ingrid Daubechies, SIAM.

5. OTHER SESSIONS

5.1 *TUTORIALS:: No tutorials

5.2 *LABORATORY::Yes (2 hrs.)

5.3 *PROJECT::Nil

6. ASSESSMENT (indicative only)

6.1 HA:: 0

6.2 QUIZZES-HA:: 10%

6.3 PERIODICAL EXAMS:: 30%

6.4 *PROJECT:: 20%

6.5 FINAL EXAM:: 40%

7. OUTCOME OF THE COURSE::After attending this course the student will be able to understand the concepts of wavelets and their applications to various disciplines.

8. *EXPECTED ENROLLMENT FOR THE COURSE::40

9. *DEPARTMENTS OTHER THAN YOUR OWN TO WHICH THIS COURSE WOULD BE OF INTEREST::

10. *ANY OTHER REMARKS:: nil